

List of Tables



2.1	The unweighted initial abundances by mass of our models, unless otherwise stated.	18
3.1	The five major stages of the evolution of our binary.	52
3.2	The key evolutionary parameters of both stars and the binary configuration at five critical points: the initial parameters, and the beginning and ends of both epochs of RLOF.	62
4.1	The populations under study.	71
4.2	The systems we use in our study.	80
5.1	Search parameters for the six close-contact binaries compiled by Abdul-Masih et al. (2022) , as used in our fitting to the BPASS v2.2.1 grid.	95
5.2	Matches from the HOKI (Stevance et al., 2020) software.	99
5.3	Matches from the TUI code.	100
5.4	The mean post-secondary-supernova parameters for the bound remnants of our binary models.	103
5.5	The matches in primary mass / primary radius to our overcontact models.	107
5.6	The observed vs. matched parameters for each of the 14 overcontact matches we find.	111
5.7	The match results with associated mean ages at match.	112
6.1	The runtimes for a grid of STARS models on a 72-core cluster, assuming each model takes 20 min/40 min/60 min respectively.	121
A.1	The mass loss rate constants for interpolation around the bi-stability jump.	140
B.1	The code versions we use in this thesis. *TUI was described in earlier papers, but Ghodla et al. (2023) was the first to name it.	144
B.2	Our general initial conditions.	145
D.1	Matches from the TUI code.	153
E.1	The remnants generated by each model from the BPASS release.	176

